

TEODORO D. LEITE

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EDUCATION

Boston University, College of Engineering

Boston, MA

B.S. Mechanical Engineering | Concentration: Aerospace Engineering

Expected May 2028

Relevant Coursework: Differential Equations, Engineering Mechanics I, CAD and Machine Components, Electric Circuits, Computational Linear Algebra, Probability & Statistics, and Physics I & II

In Progress (Fall 2026): Fluid Mechanics, Energy & Thermodynamics, Mechanics of Materials, Manufacturing Processes, Electromechanical Design

EXPERIENCE

Robosys Automation – Software Development Engineering Intern

Jun 2026 – Present

Machine Learning / Computer Vision | VoyagerAI Maritime Autonomy Program

- Develop a camera-based vessel detection and positioning pipeline for a maritime navigation product, written in C#/.NET and targeting an NVIDIA Jetson edge computer
- Train YOLOv8 vessel-detection models in Roboflow across multiple maritime datasets, and build Python tooling to convert, deduplicate, and label tens of thousands of images
- Implement the navigation math layer that converts detections to true bearing and range and outputs NMEA TTM target messages over UDP to an ARPA display
- Design and build an RC data-collection boat — an inflatable dinghy retrofit with dual electric thrusters on differential steering and a 4-camera GMSL rig — to capture real-world maritime imagery for training the detection models

Engineers Without Borders – BU Chapter – Filtration Team Technical Lead

Jan 2026 – Present

Boston, MA

- Lead design and development of water filtration systems for borehole projects serving underserved communities
- Coordinate technical learning sessions and team meetings on filtration methods, fluid systems, and water quality
- Collaborate with multidisciplinary teams to evaluate sustainable and cost-effective engineering solutions
- Apply fluid mechanics and environmental engineering principles to real-world water access challenges

PROJECTS

1984 Pontiac Firebird – Custom Temperature-Controlled Fan System

Jun 2025 – Aug 2025

Personal Project | Fall River, MA

- Designed and built an automated engine cooling system using an LM35 temperature sensor, LM393 comparator IC, and relay module to trigger cooling fans based on real-time engine temperature
- Generated Gerber files and ordered a professionally manufactured PCB through JLCPCB, replacing the breadboard prototype with a durable production-quality board
- Integrated the custom PCB into the Firebird's 12V electrical system with fuse protection, modernizing the car's cooling reliability and reducing unnecessary power draw

Intro to Engineering Design – Mechanical Game Project – Lead Designer

Jan 2026 – Present

Boston University | Boston, MA

- Designed and 3D-printed a custom enclosure for an interactive arcade-style game, modeling all parts and assemblies in SolidWorks for fit and manufacturability
- Iterated designs based on mechanical constraints, tolerances, and manufacturability across multiple prototyping cycles
- Built the electromechanical system on an Arduino with a stepper motor, joystick, arcade buttons, an LCD display, and audio output, wiring and programming the full control loop

SKILLS

Hardware & Prototyping: Circuit design, PCB fabrication (JLCPCB/Gerber), Soldering, Embedded systems (Arduino), Relay switching, Mechanical design, 3D printing, Machine shop access (BU)

Software: MATLAB, SolidWorks/CAD, Python, C++, C#/.NET, HTML, Microsoft Office

Machine Learning: Computer vision, YOLOv8 object detection, Roboflow, PyTorch/ONNX, dataset labeling & augmentation

Languages: English (fluent), Portuguese (native/bilingual)